Cookbook

Six common edits to a SWAT+ project that already runs. Each recipe names the file and the values to change, and what to check after the run. The input reference pages linked from each recipe give the full rules.

Simulation period

Edit `time.sim`:

day_start	yrc_start	day_end	yrc_end	step
0	1975	0	2020	0

- `yrc_start` and `yrc_end` are the first and last calendar years.
- `day_start` is the day of the year the run starts; 0 is read as 1, 1 January.
- `day_end` is the day of the year the run ends in the last year; 0 means 31 December.

To leave warm-up years out of the outputs, keep `time.sim` and set `nyskip` on line 3 of `print.prt` to the number of years to skip. The skipped years are still simulated.

Days outside the measured weather records are generated from `weather-wgn.cli` without a message, so a period longer than the records still runs. Compare the period with the first and last dates in the `.pcp` and `.tmp` files before reading the results.

References: time.sim, print.prt, weather outside the observed record.

Daily or monthly output

Each object row of `print.prt` has four switches, for daily, monthly, yearly and average annual files. Set the column to `y` to write that file, for example daily basin water balance:

objects	daily	monthly	yearly	avann
basin_wb	y	n	y	y

How the rows are read depends on the second value on line 7 (`use_obj_lbls`, labelled `dbout` in files from older Editors):

- `y`: rows are matched by name. A row can be added, removed or moved, and the carbon and gwflow rows can be used.
- `n`: rows are read by position in a fixed order. Do not add, remove or reorder rows: every later row would apply to the wrong output, without a message.

Daily output of many objects over a long period can take several gigabytes, so check the free disk space before a long run. After the run, `files_out.out` lists every output file that was opened.

References: print.prt, Output naming convention.

PET method

Edit `pet`, the third value in `codes.bsn`:

pet	Method	Weather it uses
0	Priestley-Taylor	temperature and solar radiation
1	Penman-Monteith	temperature, solar radiation, relative humidity and wind
2	Hargreaves	temperature

pet	Method	Weather it uses
3	weather station PET	in 62.0.1 this gives Hargreaves PET (see below)

The PET method does not change which weather files are read. A variable that the station takes from the weather generator (**sim** in weather-sta.cli) is generated in every case, so Penman-Monteith with generated humidity and wind uses generated values. With **pet** = 3, 62.0.1 replaces the measured station PET with Hargreaves PET before the HRUs use it, so this code gives Hargreaves PET times **pet_co**.

After the run, compare **pet** and **et** in **hru_wb_aa.txt** or **basin_wb_aa.txt** with the earlier run.

References: PET method, weather-sta.cli.

Green-Ampt infiltration

Three files change:

1. codes.bsn: set **gampt**, the 21st value, to 1.
2. time.sim: set **step** to the number of time steps per day, for example 24 for hourly steps.
3. The precipitation files: give sub-daily records with the same number of values per day as **step**. With **step** above 1, 62.0.1 does not split a measured daily record over the time steps, so a daily **.pcp** file gives no rain in any time step.

Plant growth, soil water and nutrient cycling still run once a day.

After the run, compare **surq_gen** and **perc** in **basin_wb_aa.txt** with the curve number run.

References: runoff method, time step, *.pcp.

HRU addition

An HRU needs a row in several files. Copy the rows of an existing HRU and change them:

1. hru.con: add a row with the next **id**. Rows must be numbered 1, 2, 3 and so on in file order. Set **area**, **lat**, **elev**, the **hru** column (the row number of the new HRU in hru-data.hru) and **wst** (a station name in weather-sta.cli).
2. hru-data.hru: add a row whose **topo**, **hydro**, **soil**, **lu_mgt**, **soil_plant_init**, **snow** and **field** names exist in the corresponding files. A misspelled name is not reported: the HRU uses the first record of that file.
3. The HRU's outflow: either give it outflow connections in hru.con (**out_tot** and the groups after it), or add it as an element of a routing unit in rout_unit.ele and the element list in rout_unit.def, as in Osu_1hru. With neither, its water leaves the model.
4. ls_unit.ele and ls_unit.def, if the project has them: add the HRU as an element. The basin output files (**basin_wb** and the others) are added up from these elements with their **bsn_frac** values as given, so the **bsn_frac** values should add up to 1 after the change.
5. object.cnt: add 1 to **hru** and to **obj**.

After the run, **checker.out** has one row per HRU, in HRU order, with the soil and hydrology values the new HRU ended up with. Check **diagnostics.out** for names that were not found, and remember that the **topo**, **hydro**, **soil**, **lu_mgt**, **soil_plant_init** and **snow** names are not checked there.

References: *.con, hru-data.hru, object.cnt, print.prt.

Carbon model

The dynamic soil carbon model needs three changes:

1. codes.bsn: set **carbon**, the 14th value, to 2. The value 1 is reserved for C-FARM, which is not implemented, and runs the model with no organic matter cycling at all. Datasets from before revision 62 used 1 for the dynamic model and must be changed.
2. file.cio: name the carbon file as the third name on the **basin** line, for example **basin codes.bsn parameters.bsn carbon.bsn**.

3. Provide carbon.bsn and carbon_lyr.bsn in the project folder. refdata/Ames_sub1 has both, and they can be copied as a start. SWAT+ stops with an error if either is missing. The old carb_coefs.cbn is no longer read.

For carbon output, print.prt must use use_obj_lbls = y, and the carbon rows are added by name. The newer rows (hru_cb_gl, hru_cb_lyr, hru_cb_cpool and others) each switch one family of files, and the legacy rows hru_cb and hru_cb_vars switch the files the carbon notebook reads. Daily per-layer carbon output is large: in Ames_sub1 it makes up most of the 9 GB of a full run, so start with a short period.

After the run, files_out.out lists the carbon files that were opened, and Carbon outputs describes their columns.

References: carbon model, carbon.bsn, older file.cio files, carb_coefs.cbn.

Sources

- Simulation period: time_read.f90 sets time%day_start to 1 when it is 0 or less.
- Output rows: basin_print_codes_read.f90 reads the second value of line 7 into pco%use_obj_labels and the object rows by position or by name accordingly.
- codes.bsn: basin_read_cc.f90. Column positions: pet 3 (bsn_cc%pet), carbon 14 (bsn_cc%cswat), gampt 21 (bsn_cc%gampt). PET with pet = 3 and sub-daily precipitation: see the Sources sections of codes.bsn and time.sim.
- HRUs: connection rows are read by hyd_read_connect.f90, HRU rows and their name lookups by hru_read.f90, counts by basin_read_objs.f90, and checker.out is written by proc_hru.f90. Basin output weighting by bsn_frac: see nutrient balance and losses.
- Carbon: the basin line of file.cio is read in readcio_read.f90, and carbon.bsn and carbon_lyr.bsn by carbon_bsn_read.f90.